

Redistributive Justice, Job Insecurity, and Perceived Technological Risk: How Labor Market Position Shapes the Legitimacy of the Welfare State in 27 OECD Countries

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Abstract

Digital technologies are transforming the standard employment relationship around which social protection was built, yet far less is known about how workers' labour-market position and their anticipation of technological change shape the perceived justice of their exchange with the welfare state. Drawing on the 2024 wave of the OECD Risks That Matter survey (19,669 respondents in 27 countries), this article examines whether perceived redistributive justice, the judgement that one receives a fair share given what one contributes, is anchored in type of employment, job insecurity, technological perceptions and the perceived deservingness of others, and whether welfare regimes condition these relationships. Preregistered hypotheses are tested with multilevel cumulative-link models with random country intercepts and standardised macroeconomic controls. Three individual-level expectations hold: concern about losing one's job or income and the perception that others receive undeserved benefits depress perceived justice, while perceiving technology as enabling raises it. Two do not: platform workers judge their exchange as more just rather than less, and perceived technological displacement risk increases rather than erodes perceived justice. At the country level, perceived justice is higher in richer countries and, net of this composition, lower in conservative than in liberal regimes; the moderation analyses further reveal a consistent institutional patterning: every regime interaction is negative, the platform and displacement associations are steepest under liberal arrangements and flattest under social-democratic ones, self-employment becomes consequential only in social-democratic regimes, and deservingness judgements weigh most heavily in conservative regimes. The findings underline the analytical distinction between redistributive preferences and justice perceptions, and recast the regime effect as one of decoupling: universalist institutions weaken the translation of platform work and technological perception into judgements of justice, whereas residual arrangements bind them together. These results bear on the European agenda on access to social protection for non-standard and platform workers: extending coverage raises what is at stake in the exchange, and its legitimacy returns will depend on entitlements being proportionate and legible.

Keywords: Redistributive justice, Labor market position, Welfare regimes, platform work, technological change

1 Introduction

Contemporary technological transformations, such as automation, artificial intelligence, algorithmic management and intermediation through digital platforms, are altering the conditions under which workers

relate to social protection systems, the subjective experience of their position in the labour market and, with it, their assessment of the justice of their relationship with the welfare state (Busemeyer and Sahm, 2022; Frey and Osborne, 2017; Gallego and Kurer, 2022; Vallas and Schor, 2020). These systems were built on the premise of stable, full-time and long-term employment, in which contributions flowed predictably from wages and rights accumulated over continuous careers (Esping-Andersen, 1993; Korpi and Palme, 1998). The progressive decoupling between that model and the actual forms of work, mainly through temporary contracts, self-employment and platform-mediated work, raises the question of how citizens today perceive the justice of the exchange they maintain with the institutions that ought to protect them. In Europe, this question has also entered the policy agenda: access to social protection for self-employed and platform workers has been the object of a Council Recommendation (Council of the European Union, 2019) and, more recently, of a directive on improving working conditions in platform work (European Parliament and Council of the European Union, 2024).

Research on attitudes towards the welfare state has examined extensively the determinants of support for redistribution, showing that exposure to labour market risk and to automation tends to increase the demand for social protection (Busemeyer and Sahm, 2022; Rehm, 2009; Thewissen and Rueda, 2019). Much less attention has been paid, by contrast, to perceived redistributive justice, that is, to the evaluative judgement of whether the benefits that an individual receives from the state constitute just compensation for the taxes and contributions they pay (Roosma et al., 2016).

This study addresses whether perceiving the functioning of the state as just is related to labour market position and the perception of technological risk. Drawing on data from the Risks That Matter survey of the OECD (2024), which covers more than 27,000 people in 27 countries, this research places European welfare states in comparative perspective and simultaneously introduces three little-explored elements. First, it distinguishes labour market position according to type of employment —formal employees, self-employed workers and platform workers—, forms that maintain unequal links with the social protection system and that have rarely been examined together in relation to perceived redistributive justice. Second, it incorporates the perception of technological risk as a source of redistributive dissatisfaction distinct from type of employment and from conventional job insecurity, distinguishing its threatening dimension —the likelihood of being displaced by technology— from its enabling dimension —the perception that technology can improve working conditions—. Third, it incorporates the perceived deservingness of others —the judgement of whether others receive benefits from the state that they do not deserve— as a predictor of the assessment that each individual makes of their own relationship with the system (Van Oorschot, 2000, 2006).

The study makes three contributions. First, it extends comparative research on welfare attitudes by situating the structural conditions of the labour market as explanatory variables of perceived redistributive justice, linking subjective assessments of the welfare state with the organisation of work (Emmenegger et al., 2012; Kalleberg, 2011). Second, it provides comparative evidence on the effect of the perception of technological risk in an attitudinal domain distinct from the one that recent literature has concentrated on, concerned above all with support for redistribution or for social investment (Busemeyer and Sahm, 2022; Thewissen and Rueda, 2019). Third, it places these processes within a comparative framework

of welfare regimes, which makes it possible to assess how different institutional contexts moderate the relationship between labour market vulnerability, technological perception and judgements of justice (Esping-Andersen, 1993; Larsen, 2008), and speaks to the discussion on how welfare states should adapt social protection to employment forms that depart from the standard employment relationship.

The argument organising this work is that the perception of justice in the relationship between what one contributes and what one receives is not solely a function of ideological predispositions or of abstract beliefs about inequality, but is anchored in workers' everyday experience of the labour market: the employment relationship that links them to the protection system, how secure they feel in their jobs, whether they perceive technology as a threat or an opportunity, and how they judge the deservingness of those who also receive from the state. Although the welfare state was designed for a world of stable employment, its perceived legitimacy is called into question by labour markets that increasingly depart from that foundational assumption (Howcroft et al., 2025; Standing, 2014).

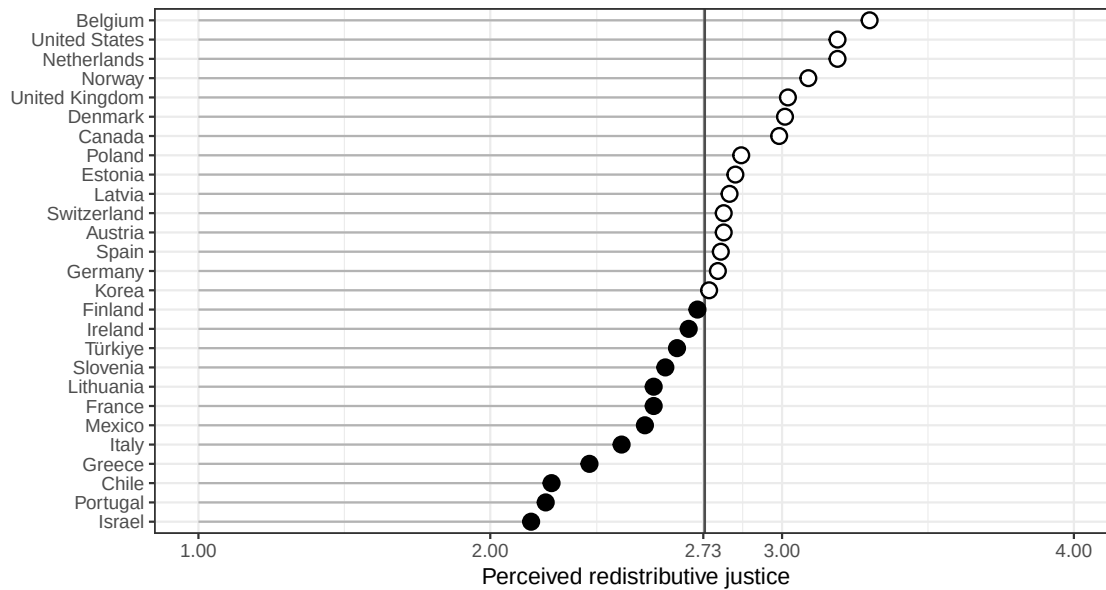
1.1 Redistributive justice as reciprocity

Research on attitudes towards the welfare state and inequality has distinguished between preferences -normative beliefs about how resources should be distributed- and perceptions -evaluative judgements about how the system actually works- (Castillo et al., 2022; McCall, 2013; Osberg and Smeeding, 2006). While the former have received greater attention, the latter have emerged as an object of study with distinct dynamics and consequences (Roosma et al., 2016; Svallfors, 2013). Perceived redistributive justice captures whether individuals consider just the exchange they maintain with the welfare state when paying taxes and contributions in return for access to services and transfers. Its theoretical foundation lies in the logic of reciprocity, the principle according to which the legitimacy of the system rests not only on its redistributive outcomes, but on the perceived justice of the institutions that allocate contributions and benefits (Mau, 2014; Rothstein, 1998). When individuals feel that their contributions yield adequate returns, institutional trust is sustained; when the reciprocal logic is perceived to be broken (disproportionate contributions, insufficient benefits or a system that favours some over others), legitimacy is eroded (Cavaillé, 2025), and social cohesion is affected.

In recent decades, the study of redistributive attitudes has shown that, despite the sustained rise in inequality in most advanced democracies, there has been no proportional increase in the demand for redistribution (Cavaillé, 2025; Kenworthy and McCall, 2007; Mijs, 2021). Explanations of this apparent disconnect point to how citizens perceive inequality and their own position in the distribution (García-Castro et al., 2020; García-Sánchez et al., 2020; Gimpelson and Treisman, 2018), perceptions that interact with meritocratic beliefs to shape which type of distribution is considered just (Castillo et al., 2025). Within this framework, the perceived justice of the fiscal exchange occupies a particular place in that it does not measure how much redistribution a person wants, but rather judges whether the system treats them in a just way given what they contribute.

Comparative evidence shows that these perceptions vary systematically across countries (Blekesaune, 2003; Schmidt-Catran, 2016). Figure 1a presents the average of perceived redistributive justice in 27 countries and shows considerable heterogeneity, independently of welfare regime type.

Figure 1: Mean of perceived redistributive justice by OECD country



(a) Source: own elaboration with data from the OECD, Risks That Matter survey 2024. Perceived redistributive justice corresponds to the degree of agreement with the item on receiving a fair share of public benefits given the taxes and contributions paid (scale 1–5). The vertical grey line shows the OECD mean of redistributive justice.

1.2 Work, type of employment and platform economy

The sociology of work has documented in multiple contexts the growing polarisation between formal, well-paid employment and an expanding stratum of precarious work (Gallie, 2013; Kalleberg, 2011). Precarious work—uncertain and unpredictable employment that shifts risk from employers to workers—has spread beyond the margins of the labour market to reach growing portions of the workforce (Kalleberg, 2011), and is not reduced to income insecurity, but entails the absence of the occupational identity, social rights and temporal stability that characterised the standard employment relationship (Standing, 2014). These transformations matter for the welfare state because its institutions were designed around that formal relationship, so that those who depart from it tend to maintain a weaker and more unequal link with the protection system (Emmenegger et al., 2012; Schwander and Häusermann, 2013). This structural position in the labour market shapes that material link, and also conditions attitudes towards inequality and redistribution (Lindh and McCall, 2020), an effect especially visible in contexts of marked commodification such as the Chilean one (Pérez-Ahumada et al., 2026).

Type of employment structures that relationship directly. Workers in formal waged employment contribute through registered contributions and accumulate rights in a legible way, which makes the relationship between what they contribute and what they receive traceable. Workers in non-formal employment face a more opaque and often less favourable relationship, since the self-employed tend to pay higher net contributions and remain excluded from unemployment insurance, while temporary workers accumulate interrupted contribution records that reduce their future benefits (Marx, 2014; Rueda, 2007). This institutional distance between insiders and outsiders in the labour market creates conditions for the latter to perceive that the system was not designed for people in their situation, which may translate into a less

favourable assessment of the justice of the fiscal exchange ([Schwander and Häusermann, 2013](#)).

Platform-mediated work constitutes a particularly acute form of non-formal employment. Algorithmic intermediation, the classification of workers as independent contractors and the fragmentation of tasks place these workers largely outside the formal architecture of social insurance ([Vallas and Schor, 2020](#); [Wood et al., 2019](#)). Several studies have shown that platform workers combine nominal autonomy with a high degree of external control and a fragmented or non-existent social protection ([Behrendt et al., 2019](#); [Schor, 2020](#)), a pattern that evidence for Latin America has documented in platform-mediated delivery, where classification as independent contractors shifts risk onto workers and leaves them outside social security ([Gutiérrez Crocco and Atzeni, 2022](#)). The European directive on platform work responds to this situation by introducing a legal presumption of employment and rules on algorithmic management ([European Parliament and Council of the European Union, 2024](#)). This structural position exposes these workers to a lack of protection that leads one to expect a comparatively lower perception of the justice of their relationship with the welfare state. These considerations give rise to the first hypothesis:

H1: Workers in non-formal employment —platform work and self-employment— will perceive lower redistributive justice than workers in formal waged employment.

1.3 Job insecurity

Beyond type of employment, the level of job insecurity experienced by workers constitutes a distinct dimension of vulnerability. Job insecurity is defined as the subjective assessment of the concern of losing one's own job or self-employment income in the short term, with the chronic uncertainty and anticipatory anxiety this entails ([De Witte, 2005](#); [Sverke et al., 2002](#)). Unlike objective precarity, it is a perceived state that can also affect formally protected workers and whose effects on behaviour and attitudes are widely documented ([Anderson and Pontusson, 2007](#); [Chung and Van Oorschot, 2011](#)).

From the perspective of redistributive justice, job insecurity raises what is at stake in the relationship between contributions and benefits. The literature on risk and redistribution has shown that exposure to insecurity increases support for social protection ([Cusack et al., 2006](#); [Iversen and Soskice, 2001](#); [Rehm, 2009](#)). The relationship with perceived redistributive justice may, however, operate in a different direction: workers who anticipate losing their job assess with greater scrutiny what they would receive from the state in that scenario, and when that expectation is not accompanied by confidence in the protective capacity of the system, they are likely to perceive the exchange as less just ([Marx, 2014](#)). The second hypothesis is thus proposed:

H2: Greater perceived job insecurity will be associated with a lower perception of redistributive justice.

1.4 Perceived technological change

The rise of digital technologies has introduced an additional source of occupational vulnerability, distinct from conventional job insecurity. Estimates of the proportion of jobs susceptible to automation vary widely, but even the most conservative ones anticipate significant changes in the content of tasks and in the viability of numerous occupations (Arntz et al., 2016; Autor, 2015; Bustelo et al., 2020; Frey and Osborne, 2017). What matters for welfare attitudes is the objective magnitude of that process, and also the way in which workers anticipate its effects on their own job (Gallego and Kurer, 2022). That anticipation comprises both the perception of a displacement risk (the probability of losing the job to technology) and the perception of a benefit (the probability that technology will improve the quality of work).

1.4.1 Perception of technological displacement risk

On the one hand, technological change refers to the perception of technological displacement risk, that is, the probability that one's own job will be replaced by a robot or by artificial intelligence, substituted by a provider offering a similar service through a platform, or rendered unviable by the obsolescence of one's own digital skills. Research has shown that the fear of automation is distributed unevenly across occupations and social groups and that it has measurable political and attitudinal consequences (Dekker et al., 2017; Im et al., 2019; Kurer, 2020). The literature on risk and redistribution has robustly established that perceived exposure to technological risk increases the demand for social protection (Busemeyer and Sahm, 2022; Iversen and Soskice, 2001; Rehm, 2009; Thewissen and Rueda, 2019). Following that logic, workers who perceive their job as threatened by technology hold greater expectations of protection and, when contrasting these with what they actually receive, are more likely to assess their exchange with the welfare state as insufficient (Marx, 2014). The third hypothesis is thus derived:

H3: Workers who perceive a higher probability of being displaced by automation, platform competition or the obsolescence of their skills will report a lower perception of distributive justice.

1.4.2 Perception of technological benefit

On the other hand, technological change refers to the perception of technological benefit, which addresses the probability that technology will make working hours more compatible with private life, reduce the dangerousness or physical demands of work, or lessen its monotony and mental load. This enabling dimension has received much less attention than its threatening counterpart, even though workers' orientation towards technology is not solely negative (Autor, 2015; Brougham and Haar, 2018). Unlike displacement risk, perceiving technology as an improvement in working conditions implies lower exposure to risk and greater optimism about one's occupational trajectory. To the extent that the assessment of the exchange with the welfare state reflects the individual's position of economic security (Rehm, 2009), this positive orientation can be expected to be associated with a more favourable perception of redistributive justice. The fourth hypothesis follows from this:

H4: Workers who perceive technology as enabling —reducing physical demands, improving the balance between work and private life or making work less stressful or monotonous— will perceive greater redistributive justice.

1.5 Perceived deservingness of others

Perceptions of redistributive justice do not depend solely on structural position, but also on the normative frameworks through which individuals assess who and what deserves to receive from the state. Deservingness literature holds that people direct their solidarity towards the groups they consider deserving and withdraw it from those they judge undeserving, on the basis of criteria such as control over one's own situation, attitude, reciprocity, identity and need (Petersen, 2012; Van Oorschot, 2000). In strongly commodified contexts, meritocratic and market-justice beliefs reinforce the tendency to consider undeserving those who are perceived not to contribute enough (Castillo et al., 2025). These judgements about the deservingness of others are analytically distinct from the judgement about the justice of one's own relationship with the system: a person may consider themselves a contributor who does not receive their just share and, at the same time, judge that others receive benefits they do not deserve. Both assessments are, however, part of the same moral economy of welfare, and they can be expected to be connected (Mau, 2014; Van Oorschot et al., 2017).

The connection between the two judgements operates through the norm of reciprocity. The resources that an individual contributes appear to sustain those who do not fulfil their part, which erodes the sense that the system treats them in a just way (Cavallé, 2025; Reeskens and Van Oorschot, 2013). Comparative research has shown that deservingness perceptions robustly structure support for the welfare state and that the control criterion occupies a central place in those judgements (Laenen, 2020; Van Oorschot, 2006). It can therefore be expected that those who perceive greater abuse of the system by others will assess their own relationship with it as less just. The fifth hypothesis is proposed:

H5: Workers who perceive that many people receive public benefits without deserving them will report a lower perception of redistributive justice.

1.6 Welfare regimes and institutional context

Individual perceptions of redistributive justice are formed in institutional contexts that vary systematically across welfare regimes (Arts and Gelissen, 2002; Esping-Andersen, 1993). Esping-Andersen (1993)'s classic typology distinguishes three worlds of welfare capitalism according to how they organise the relationship between state, market and family in the provision of social protection. In social democratic regimes (mainly Nordic countries), protection is comprehensive and universal, and redistribution constitutes a central expression of social solidarity sustained by universalist institutions (Korpi and Palme, 1998). In liberal regimes (mainly Anglo-Saxon countries and, for example, Chile as a case of a commodified welfare state in Latin America), protection is residual and targeted, is organised around market participation and is politically more contested Pérez-Ahumada (2023). In conservative-corporatist regimes (continental

Europe), rights are strongly tied to employment status, which preserves status differences and links protection to the employment trajectory (Ferrera, 1996).

Alongside institutional design, macroeconomic factors bear on welfare attitudes: income inequality is associated with perceptions of the justice of the system, although the direction of that relationship is a matter of debate (Finseraas, 2009; Schmidt-Catran, 2016), and the level of economic development conditions both the fiscal capacity of the state and citizens' expectations regarding social protection (Blekesaune, 2003). Taken together, the perception of redistributive justice can be expected to be, on average, higher in social democratic regimes, where protection is more comprehensive and universal, than in liberal and conservative regimes (Jæger, 2006; Larsen, 2008; Svallfors, 1997). Hence the sixth hypothesis:

H6: On average, workers in social democratic regimes will perceive higher redistributive justice than workers in liberal and conservative regimes.

The institutional context shifts the average level of perceptions, and may moderate the effect of individual predictors. The institutional distance between workers in non-formal employment and the protection system is greater in liberal and conservative regimes than in social democratic ones, which are universalist in character (Ferrera, 1996; Schwander and Häusermann, 2013). It can therefore be expected that the negative effect of labour market position on perceived redistributive justice will be more pronounced in the former:

H7: The negative relationship between non-standard employment (platform work and self-employment) and perceived redistributive justice will be moderated by the welfare regime.

An analogous expectation can be formulated for the perception of technological risk. In universalist regimes, protection against the risks associated with digitalisation is more robust and less dependent on the individual employment trajectory, which could moderate the effect of perceived displacement risk on the assessment of the fiscal exchange (Busemeyer and Sahm, 2022; Fan et al., 2024; Gallego and Kurer, 2022):

H8: The negative effect of the perception of technological displacement risk on the perception of redistributive justice will be moderated by the welfare regime.

Finally, the salience of deservingness judgements varies according to institutional design. In universalist regimes, where benefits are conceived as general rights rather than as targeted assistance, the distinctions between deserving and undeserving tend to be less pronounced than in residual regimes, organised around means testing (Larsen, 2008; Van Oorschot et al., 2017). It can therefore be expected that the effect of the perceived deservingness of others on the justice of one's own exchange will be stronger in liberal and conservative regimes:

H9: The negative effect of the perceived deservingness of others on the perception of redistributive justice will be moderated by the welfare regime.

2 Data, variables and methods

2.1 Data

This study uses data from the most recent wave (2024) of the Risks That Matter (RTM) survey of the OECD, a cross-national public opinion survey that includes 27,229 people aged 18 to 64 in 27 member countries: Austria, Belgium, Canada, Chile, Denmark, Estonia, Finland, France, Germany, Greece, Ireland, Israel, Italy, Korea, Latvia, Lithuania, Mexico, the Netherlands, Norway, Poland, Portugal, Slovenia, Spain, Switzerland, Türkiye, the United Kingdom and the United States. The survey collects nationally representative data using stratified sampling with post-stratification weights calibrated by age, sex, region and educational level. The 2024 wave is especially relevant for this study because it captures attitudes in the period following the COVID-19 pandemic, which accelerated the digitalisation of work and expanded platform employment.

2.2 Variables

The dependent variable is the perception of redistributive justice, measured by the item: “I feel that I receive a fair share of public benefits, given the taxes and social contributions that I pay and/or have paid in the past”, answered on a five-point Likert scale ranging from strongly disagree to strongly agree, and is treated as an ordinal variable in all the analyses.

Three sets of independent variables constitute the core of the analysis. The first concerns labour market position and job insecurity. Type of employment is operationalised as a three-category variable distinguishing formal waged employment (permanent or long-term dependent employment), platform work (work mediated by digital platforms) and self-employment. Job insecurity is measured by the item “Losing a job or self-employment income”, asked within the battery “Thinking about the next year or two, how concerned are you about each of the following?”, with response categories ranging from not at all concerned to very concerned.

The second set of variables captures the perception of technological change with respect to one’s own job. These variables are measured through a battery that asks: “How likely do you think it is that the following will happen in your job (or job opportunities) in the next five years?”, with items rated on a four-point probability scale. The battery includes four displacement items —replacement of the job by a robot, by an artificial intelligence tool such as ChatGPT, or by a person offering a similar service through an internet platform, and the loss of the job through not being sufficiently technologically skilled— and three benefit items —that technology will make working hours more compatible with private life, make work less dangerous or physically demanding, and make it less boring, repetitive, stressful or mentally demanding. However, the exploratory factor analysis carried out on these seven items yields a two-factor

structure: three of the four displacement items load on a first factor, the three benefit items on a second, and the item on replacement by artificial intelligence does not load cleanly on either factor, so it is excluded from the scales. Two scales are therefore built: Technological Displacement Risk (the average of the three displacement items that load together) and Perceived Technological Benefit (the average of the three benefit items). In both cases, higher values indicate a greater perception of the corresponding risk or benefit. The full exploratory factor analysis is reported in the appendix at Table 4.

The third set of independent variables concerns the perceived deservingness of others, operationalised through the item that measures the degree of agreement with the statement “Many people receive public benefits without deserving them”, rated on a five-point scale (1 = strongly disagree to 5 = strongly agree). Unlike the dependent variable, this item captures the judgement of the deservingness of other beneficiaries rather than of one’s own exchange with the system, a central distinction in the deservingness literature (Van Oorschot, 2000, 2006). Higher values indicate a stronger perception that others receive benefits without deserving them.

The welfare regime is operationalised as a three-category variable —social democratic, liberal and conservative-corporatist— following Esping-Andersen (1993) and Bamba (2007). The countries of southern Europe and of central and eastern Europe are integrated into the conservative category, given the initial specificity of the post-socialist regimes and their progressive convergence towards the continental types (Fenger, 2006; Ferrera, 1996), while the Baltic states, characterised by a more liberal orientation of their post-socialist reforms, are incorporated into the liberal regime (Aidukaite, 2011). Among the non-European countries, Chile and Mexico are classified as liberal, following the characterisation of their welfare regimes as market-centred (Martínez Franzoni, 2008), while Israel, Korea and Türkiye are assigned to the conservative category. The full classification by country, with its sources, is reported in the appendix at Table 5.

At the country level, four macroeconomic indicators are included: the Gini coefficient of disposable income and GDP per capita (PPP, constant international dollars), both drawn from the World Bank’s World Development Indicators; the unemployment rate (ILO estimate, from the same source); and public social expenditure as a percentage of GDP, drawn from the OECD Social Expenditure Database (SOCX). For each country the most recent value available within the 2015–2024 window is used, GDP per capita is log-transformed, and all four indicators are grand-mean standardised, so that their coefficients express the association with a difference of one standard deviation between countries. Income inequality and the level of economic development have been shown to condition welfare attitudes (Blekesaune, 2003; Finseraas, 2009; Schmidt-Catran, 2016), while social expenditure provides a continuous indicator of welfare effort that complements the categorical regime typology.

All the models include control variables associated with welfare attitudes and perceived redistributive justice: sex, age, educational level (highest level of studies completed) and income decile (self-reported position in the national income distribution).

A summary of these variables and their main descriptive statistics is available in Table 1; the macroeconomic indicators are reported there on their original scales, with statistics computed over individuals, so each

country weighs according to its sample size.

Table 1: Main descriptive statistics

Label	Stats / Values	Freqs (% of Valid)	Valid
Redistributive justice	1. Strongly disagree	3600 (18.3%)	19669 (100.0%)
	2. Disagree	4765 (24.2%)	
	3. Neither agree nor disagree	5641 (28.7%)	
	4. Agree	4255 (21.6%)	
	5. Strongly agree	1408 (7.2%)	
Sex	1. Man	10520 (53.5%)	19669 (100.0%)
	2. Woman	9149 (46.5%)	
Age group	1. 18-24	2136 (10.9%)	19669 (100.0%)
	2. 25-34	4626 (23.5%)	
	3. 35-44	4854 (24.7%)	
	4. 45-54	4599 (23.4%)	
	5. 55-64	3454 (17.6%)	
Educational level	1. Low	1546 (7.9%)	19669 (100.0%)
	2. Medium	8147 (41.4%)	
	3. High	9976 (50.7%)	
Equivalised disposable household income decile (2023)	Mean (sd) : 4.9 (2.9)	1 : 3398 (17.3%)	19669 (100.0%)
	min < med < max:	2 : 1844 (9.4%)	
	1 < 5 < 10	3 : 1788 (9.1%)	
	IQR (CV) : 5 (0.6)	4 : 2652 (13.5%)	
		5 : 2322 (11.8%)	
		6 : 1815 (9.2%)	
		7 : 1488 (7.6%)	
		8 : 1424 (7.2%)	
		9 : 1300 (6.6%)	
		10 : 1638 (8.3%)	
Self-employment worker	1. No	16887 (85.9%)	19669 (100.0%)
	2. Yes	2782 (14.1%)	
Platform worker	1. No	17825 (90.6%)	19669 (100.0%)
	2. Yes	1844 (9.4%)	
Lose job	1. Not at all concerned	2927 (14.9%)	19669 (100.0%)
	2. Not so concerned	5453 (27.7%)	
	3. Somewhat concerned	6110 (31.1%)	
	4. Very concerned	5179 (26.3%)	

Label	Stats / Values	Freqs (% of Valid)	Valid
Technological displacement	Mean (sd) : 2.1 (0.9) min < med < max: 1 < 2 < 4 IQR (CV) : 1.3 (0.4)	13 distinct values	19669 (100.0%)
Perceived technological benefit	Mean (sd) : 2.6 (0.8) min < med < max: 1 < 2.7 < 4 IQR (CV) : 1 (0.3)	13 distinct values	19669 (100.0%)
Deservingness of others	1. Strongly disagree 2. Disagree 3. Neither agree nor disagree 4. Agree 5. Strongly agree	624 (3.2%) 1727 (8.8%) 4434 (22.5%) 6670 (33.9%) 6214 (31.6%)	19669 (100.0%)
Welfare regime	1. Liberal 2. Social-democratic 3. Conservative	6563 (33.4%) 2957 (15.0%) 10149 (51.6%)	19669 (100.0%)
Gini coefficient (disposable income)	Mean (sd) : 32.8 (5) min < med < max: 24.7 < 32.4 < 43.7 IQR (CV) : 5 (0.2)	26 distinct values	19669 (100.0%)
GDP per capita (PPP)	Mean (sd) : 57093.7 (20396.3) min < med < max: 21970.4 < 53412.1 < 118833.3 IQR (CV) : 18634.9 (0.4)	27 distinct values	19669 (100.0%)
Unemployment rate (%)	Mean (sd) : 5.7 (2.3) min < med < max: 2.7 < 5.2 < 11.4 IQR (CV) : 3.2 (0.4)	25 distinct values	19669 (100.0%)
Social expenditure (% of GDP)	Mean (sd) : 21.5 (6) min < med < max: 10 < 23 < 31.6 IQR (CV) : 10.1 (0.3)	27 distinct values	19669 (100.0%)

2.3 Methods

The hypotheses were pre-registered on the Open Science Framework (anonymised view-only [link](#)). Statistical analyses were conducted in R version 4.3.2.

Multilevel ordinal logistic regression models with individuals nested in countries are used to examine the relationship between labour market position, the perception of technological risk or benefit, the deservingness of others and perceived redistributive justice. This approach accounts for the hierarchical structure of the comparative data and decomposes the variance of the outcome between individual and country characteristics (Hox et al., 2017; Snijders and Bosker, 2012), with random intercepts estimated for each country. The models are estimated sequentially at the individual level: type of employment (Model 1), adding job insecurity (Model 2), the perceptions of technological risk and benefit (Model 3), the perceived deservingness of others (Model 4) and the sociodemographic controls (Model 5); the country level is then incorporated with the macroeconomic indicators (Models 6 and 7) and the type of welfare regime (Model 8).

Model 9 introduces the interaction between the labour market predictors, and Models 10 to 14 add cross-level interactions with the type of welfare regime to test the moderation hypotheses. All the analyses use the individual survey weights provided by the OECD. The models are estimated with the `clmm` function of the `ordinal` package in R.

3 Analysis

Figure 2 presents the correlation matrix of the main variables, with coefficients ranging from low to moderate. The strongest associations of perceived redistributive justice are with being a platform worker ($r = 0.32$, $p < 0.001$), perceived technological benefit ($r = 0.26$, $p < 0.001$) and technological displacement ($r = 0.24$, $p < 0.001$), all of them positive and moderate. Deservingness of others, in contrast, shows a negative association ($r = -0.16$, $p < 0.01$).

Among the independent variables, concern about job and income loss correlates with being a platform worker ($r = 0.15$, $p < 0.001$) and with the two technological perceptions ($r = 0.30$ with displacement; $r = 0.17$ with benefit). Technological displacement and perceived benefit are themselves positively correlated ($r = 0.48$), contrary to the initial intuition that risk and benefit perceptions would be negatively related.

3.1 Multilevel ordinal logistic regression models

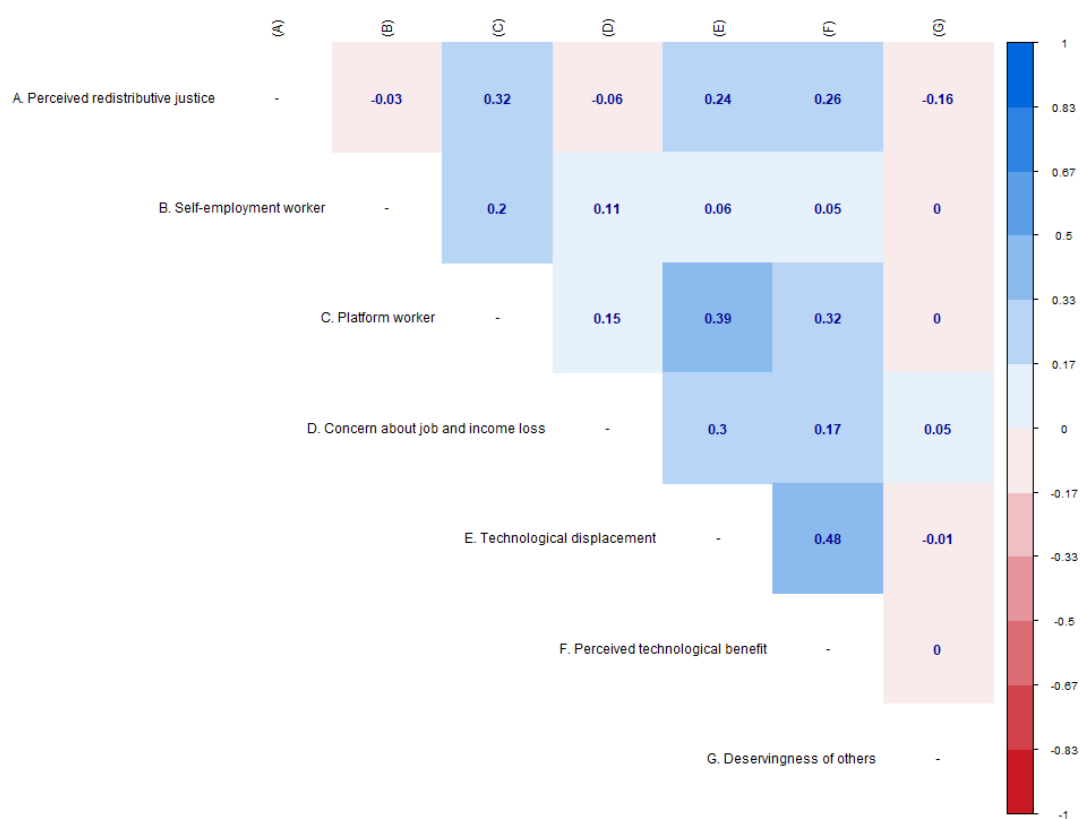
Table 2: Multilevel models for perceived redistributive justice

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Self-employed worker (ref. No)	-0.059 (0.038)	-0.056 (0.038)	-0.055 (0.039)	-0.066 (0.039)	-0.067 (0.039)	-0.064 (0.039)	-0.064 (0.039)	-0.063 (0.039)
Platform worker (ref. No)	1.177 *** (0.049)	1.193 *** (0.049)	0.838 *** (0.050)	0.854 *** (0.050)	0.788 *** (0.051)	0.788 *** (0.051)	0.788 *** (0.051)	0.787 *** (0.051)
Concern about job and income loss		-0.058 ***	-0.180 ***	-0.175 ***	-0.164 ***	-0.163 ***	-0.163 ***	-0.163 **

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
		(0.014)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Technological displacement			0.372 ***	0.366 ***	0.347 ***	0.348 ***	0.348 ***	0.348 ***
			(0.020)	(0.020)	(0.020)	(0.020)	(0.020)	(0.020)
Perceived technological benefit			0.463 ***	0.465 ***	0.438 ***	0.438 ***	0.438 ***	0.439 ***
			(0.021)	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)
Deservingness of others				-0.183 ***	-0.179 ***	-0.179 ***	-0.179 ***	-0.178 **
				(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
Gini coefficient (standardised)						-0.135	-0.095	-0.086
						(0.087)	(0.101)	(0.097)
GDP per capita (standardised)						0.217 *	0.206 *	0.191 *
						(0.087)	(0.090)	(0.084)
Unemployment rate (standardised)							-0.032	-0.047
							(0.081)	(0.075)
Social expenditure (standardised)							0.071	0.150
							(0.093)	(0.093)
<i>Welfare regime (Ref.= Liberal)</i>								
Social-democratic								-0.154
								(0.246)
Conservative								-0.350 *
								(0.165)
BIC	51477.596	51470.595	50025.832	49857.231	49766.775	49772.518	49791.434	49806.53
Numb. obs.	17296	17296	17296	17296	17296	17296	17296	17296
Groups (Country)	27	27	27	27	27	27	27	27
Variance: Country: (Intercept)	0.238	0.229	0.246	0.229	0.227	0.135	0.132	0.111

Note: Cells contain regression coefficients with standard errors in parentheses. * p < 0.05; ** p < 0.01; *** p < 0.001. Models 5 to 8 control for sex, age, education and income; sex, age and income significant effects, education does not.

Figure 2: Correlation matrix of the main variables



Source: own elaboration with data from the OECD, Risks That Matter survey 2024

Cumulative-link mixed models were fitted for perceived redistributive justice, with a random intercept for country. Note that the ordinal package reports the number of observations as the sum of the sampling weights rather than the count of cases (Haubo and Christensen, 2018); the models were therefore estimated on 19,669 respondents nested in 27 countries (weighted $N = 17,296$). Coefficients are cumulative log-odds: a positive value indicates greater odds of reporting a higher category of perceived redistributive justice.

Table 2 follows a block-building logic at the individual level — type of employment (Model 1), concern about job and income loss (Model 2), the technological perceptions (Model 3), deservingness of others (Model 4) and sociodemographic controls (Model 5) — before turning to the country level: Gini coefficient and GDP per capita (Model 6), unemployment rate and social expenditure (Model 7) and welfare regime (Model 8), with all macro indicators standardised. Between the first and the final model the country-level variance declines from 0.238 to 0.111, with the sharpest drop when the macroeconomic indicators enter (0.227 to 0.135 in Model 6). The BIC reaches its minimum in Model 5 (49,766.8) and rises slightly across the macro blocks, indicating that the country-level indicators absorb between-country variance without improving overall fit.

Platform work is the strongest and most stable predictor. In Model 1 its coefficient is 1.177 ($p < 0.001$; $OR \approx 3.2$); it attenuates as the technological and deservingness blocks enter but remains large in the fully adjusted Model 8 (0.787, $p < 0.001$; $OR \approx 2.2$), so platform workers have roughly twice the odds of expressing higher perceived redistributive justice. Self-employment is non-significant throughout (≈ -0.06). Concern about job and income loss is negative and strengthens once technological perceptions are held constant, moving from -0.058 in Model 2 to -0.163 in Model 8 ($p < 0.001$).

The technological perceptions (from Model 3) are uniformly positive. Perceived technological benefit carries the largest effect (≈ 0.44 – 0.47 , $p < 0.001$), followed by technological displacement (≈ 0.35 – 0.37 , $p < 0.001$). The positive sign on technological displacement runs counter to the hypothesised direction, a point developed further in the discussion. Deservingness of others (Model 4) is negative and stable (≈ -0.18 , $p < 0.001$), and the addition of sociodemographic controls in Model 5 leaves this individual-level structure essentially unchanged.

At the country level, GDP per capita is the only macroeconomic indicator with a significant association: perceived redistributive justice is higher in richer countries (0.217, $p < 0.05$ in Model 6; 0.191, $p < 0.05$ in Model 8), and its inclusion absorbs a substantial share of the between-country variance. The Gini coefficient, the unemployment rate and social expenditure are not significant. Welfare regime, entered in Model 8 net of these indicators, shows one significant contrast: perceived justice is lower in conservative regimes than in liberal ones (-0.350 , $p < 0.05$), while the social-democratic contrast is non-significant (-0.154).

Table 3: Multilevel interaction models for perceived redistributive justice

	Model 9	Model 10	Model 11	Model 12	Model 13	Model 14
Self-employed worker (ref. No)	-0.018	0.013	-0.065	-0.062	-0.064	-0.065

	Model 9	Model 10	Model 11	Model 12	Model 13	Model 14
	(0.041)	(0.068)	(0.039)	(0.039)	(0.039)	(0.039)
Platform worker (ref. No)	0.873 ***	0.788 ***	0.957 ***	0.783 ***	0.786 ***	0.786 ***
	(0.057)	(0.051)	(0.078)	(0.051)	(0.051)	(0.051)
Concern about job and income loss	-0.164 ***	-0.163 ***	-0.163 ***	-0.096 ***	-0.162 ***	-0.163 ***
	(0.015)	(0.015)	(0.015)	(0.025)	(0.015)	(0.015)
Technological displacement	0.349 ***	0.350 ***	0.349 ***	0.349 ***	0.455 ***	0.346 ***
	(0.020)	(0.020)	(0.020)	(0.020)	(0.030)	(0.020)
Deservingness of others	-0.178 ***	-0.179 ***	-0.179 ***	-0.178 ***	-0.180 ***	-0.129 ***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.023)
Social-democratic	-0.157	-0.108	-0.092	0.012	0.381	0.057
	(0.246)	(0.247)	(0.245)	(0.268)	(0.266)	(0.287)
Conservative	-0.353 *	-0.345 *	-0.334 *	-0.034	-0.054	-0.044
	(0.165)	(0.166)	(0.165)	(0.186)	(0.184)	(0.200)
Self-employed X Platform worker	-0.368 **					
	(0.115)					
Self-employed X Social-democratic		-0.418 ***				
		(0.126)				
Self-employed X Conservative		-0.039				
		(0.086)				
Platform worker X Social-democratic			-0.596 ***			
			(0.134)			
Platform worker X Conservative			-0.136			
			(0.108)			
Job and income loss X Social-democratic				-0.060		
				(0.044)		
Job and income loss X Conservative				-0.115 ***		

	Model 9	Model 10	Model 11	Model 12	Model 13	Model 14
				(0.032)		
Technological displacement X					-0.258 ***	
Social-democratic					(0.050)	
Technological displacement X Conservative					-0.136 ***	
					(0.038)	
Deservingness X Social-democratic						-0.058
						(0.041)
Deservingness X Conservative						-0.081 **
						(0.030)
BIC	49806.029	49813.878	49805.971	49812.848	49797.261	49818.774
Numb. obs.	17296	17296	17296	17296	17296	17296
Groups (Country)	27	27	27	27	27	27
Variance: Country: (Intercept)	0.110	0.111	0.110	0.109	0.110	0.111

Note: Cells contain regression coefficients with standard errors in parentheses. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. All models control for sex, age, education and income; sex, age and income show significant effects, education does not.

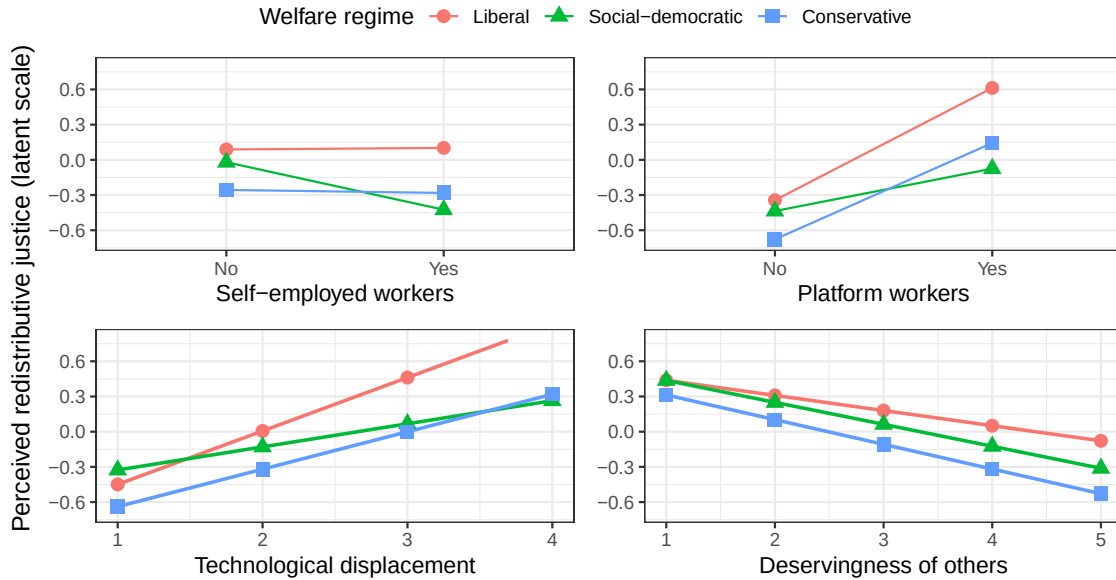
Table 3 takes the fully adjusted specification and introduces one interaction at a time: self-employed $\times$ platform (Model 9), followed by the cross-level products of welfare regime with self-employment (Model 10), platform work (Model 11), concern about job and income loss (Model 12), technological displacement (Model 13) and deservingness (Model 14). The BIC varies little across these models (49,797.3–49,818.8) and the country variance is stable (≈ 0.11). The four regime moderations are displayed in Figure 3a; because the emmeans package (Lenth and Piaskowski, 2017) evaluates the fitted model on the latent scale, the panels show predicted latent propensity rather than category-specific probabilities, which is why each moderation reduces to a single linear panel.

The individual-level interaction is negative —self-employed $\times$ platform = -0.368 ($p < 0.01$)— so the platform effect is weaker among workers who are simultaneously self-employed. The cross-level terms then qualify the regime contrasts of Table 2. Self-employment matters only in social-democratic regimes ($\times$ social-democratic = -0.418 , $p < 0.001$; $\times$ conservative n.s.), shown in the top-left panel as the steep social-democratic decline against flat liberal and conservative lines. Platform work is positive across all regimes (top right), but its slope is much shallower under social-democratic arrangements ($\times$ social-democratic = -0.596 , $p < 0.001$; $\times$ conservative n.s.).

The remaining interaction terms are also negative. Technological displacement is positive in every regime (bottom left) but significantly attenuated relative to liberal regimes ($\times$ social-democratic = -0.258 ; $\times$ conservative = -0.136 ; both $p < 0.001$). Deservingness is negative throughout (bottom right), and here the negative interaction implies amplification rather than attenuation: the slope is steepest in conservative regimes ($\times$ conservative = -0.081 , $p < 0.01$), with a non-significant social-democratic term. Concern about job and income loss interacts significantly only with the conservative contrast (-0.115 , $p < 0.001$). The conservative main effect, significant in Models 9 to 11 (≈ -0.33 to -0.35 , $p < 0.05$), loses significance in Models 12 to 14, where it refers to the contrast at the zero point of the interacted predictor rather than to an average difference.

A consistent pattern emerges: every regime interaction with an individual-level predictor is negative, so each association is weaker—or more adverse—than under liberal arrangements. For platform work and technological displacement this translates into visibly flatter slopes in social-democratic regimes, consistent with the view that encompassing welfare institutions partially decouple distributive attitudes from individual technological exposure. The exceptions to this dampening logic are self-employment, which becomes consequential—and negative—only under social-democratic arrangements, and deservingness, whose negative gradient is steepest in conservative regimes.

Figure 3: Estimated latent propensity for perceived redistributive justice, by selected predictors across welfare regimes



(a) Source: own elaboration with data from the OECD, Risks That Matter survey 2024.

4 Discussion

This article asked whether the perceived justice of the fiscal exchange between citizens and the state is anchored in labour-market position and in how workers anticipate technological change, and whether welfare institutions condition that relationship. The results lend qualified support to the reciprocity

framework that motivated the analysis, but their main interest lies where they depart from the risk-and-redistribution tradition from which the hypotheses were drawn. Three individual-level expectations held: perceived job insecurity and the perception that others receive undeserved benefits both depressed perceived redistributive justice (H2, H5), while perceiving technology as enabling raised it (H4), the largest effect among the technological perceptions. Two did not: non-standard employment was not associated with lower perceived redistributive justice, and platform workers reported it more strongly (H1); and perceived technological displacement risk, expected to erode justice, instead increased it (H3). At the country level, perceived justice proved higher in richer countries; the moderation analyses, finally, disclosed an institutional patterning that regime main effects only partly reveal.

The confirmed hypotheses support the moral-economy reading on which the study rests. Workers concerned about losing their job or income over the coming year or two judge the exchange as less just (H2), and those who perceive others as drawing unearned benefits extend that judgement to their own relationship with the system (H5), consistent with the view that lateral deservingness assessments and self-referential justice belong to a single moral economy of welfare (Mau, 2014; Van Oorschot, 2006). That the enabling dimension of technology carries the largest effect (H4) is a reminder that workers' orientation towards technology is not solely defensive, and that its optimistic pole, long neglected relative to its threatening counterpart, matters for welfare attitudes (Autor, 2015; Brougham and Haar, 2018).

Self-employment was unrelated to perceived redistributive justice throughout, which is unsurprising given a category ranging from established professionals to precarious own-account workers whose contrasting ties to the protection system plausibly offset one another. The reversal for platform work — roughly twice the odds of higher perceived justice ($OR \approx 2.2$) — admits two readings. Compositionally, self-reported platform work in OECD samples is heterogeneous, capturing supplementary and intermittent earners alongside the platform-dependent (Schor, 2020; Vallas and Schor, 2020); the negative self-employed $\times$ platform interaction localises the effect among workers most likely to combine platform earnings with other employment. Substantively, perceived redistributive justice is a judgement of proportionality, not of absolute generosity: workers loosely attached to the contributory system contribute and expect little, so a thin return reads as proportionate, whereas formal employees with visible contributions and legible entitlements may by the same token feel short-changed (Schwander and Häusermann, 2013).

The positive association between perceived technological displacement risk and perceived redistributive justice (H3) is the more consequential departure, and its failure to follow a strong prior is itself informative. Conceptually, the dependent variable is an evaluation of the justice of one's own exchange, not a demand for redistribution. The literature behind H3 establishes that perceived risk raises the demand for protection (Busemeyer and Sahm, 2022; Iversen and Soskice, 2001; Thewissen and Rueda, 2019), but wanting more redistribution and judging the present exchange as just are analytically distinct and need not move together; a worker may want the state to do more in the face of automation while still considering that, given what they pay, they receive a just share. The reversal therefore partly reflects measuring a different attitudinal object, and confirms the preferences–perceptions distinction the article insists on (Roosma et al., 2016). Compositionally, the diffusion of generative artificial intelligence has extended perceived exposure into formal, higher-skill and digitally engaged workers (Gallego and Kurer, 2022) — those with the most

continuous link to the contributory system — so that perceived risk co-travels with relative inclusion rather than marginality.

The moderation evidence favours this reading. Were the positive coefficient produced by institutional reassurance, it should be largest where protection is most encompassing; instead it is steepest in liberal regimes and significantly attenuated in social-democratic and conservative ones (H8). A temporal contrast reinforces the point: the displacement battery is prospective and abstract, asking about a five-year horizon, whereas the job-insecurity item that behaves as predicted (H2) concerns concrete income loss over the next year or two. Near-term, tangible insecurity erodes justice; distant technological prognostication does not, and may instead index a reflective engagement with the future of work bound up with other favourable dispositions (Dekker et al., 2017; Kurer, 2020). A measurement caveat sharpens rather than dissolves this: displacement and benefit are positively correlated with each other ($r = 0.48$) and with the outcome, and although the factor analysis separates them, the shared sign suggests a common component of engagement with the technological frame, which future instrumentation should separate from valence.

The comparative results recast the regime argument. The strongest country-level patterning is economic rather than institutional: perceived redistributive justice is higher in richer countries, GDP per capita absorbs a substantial share of the between-country variance, and inequality, unemployment and social expenditure show no association (Blekesaune, 2003). Net of this composition, regime type orders perceived justice only partially (H6): the conservative contrast is negative and significant relative to liberal regimes, while the social-democratic one is indistinguishable — a pattern at odds with the classic regime hypothesis and consistent with the heterogeneity already visible in Figure 1a. Every regime interaction is negative, though the pattern only partly matches the universalist–residual expectation: the positive association of platform work is steepest under liberal arrangements and flattest under social-democratic ones, whereas the self-employment effect, predicted to be strongest in residual regimes, emerges only under social-democratic arrangements (H7); the displacement effect is dampened in both non-liberal regimes (H8); and the deservingness effect is steepest in conservative regimes and indistinguishable between liberal and social-democratic ones (H9). For platform work and the technological perceptions, the reading is that universalist institutions decouple perceived redistributive justice from individual labour-market exposure, whereas residual, market-tied arrangements bind the two together (Larsen, 2008; Rothstein, 1998). The self-employment exception is consistent with the same reciprocity logic once institutional distance is understood in relative terms: where dependent employment is most generously covered, own-account workers stand furthest from the prevailing standard of protection, and their exchange with the state reads as comparatively less just (Schwander and Häusermann, 2013). This refines the institutional-logic tradition: regimes act less by raising the level of satisfaction than by governing how far it fragments along individual lines, a dynamic especially salient in highly commodified contexts such as the Chilean one (Emmenegger et al., 2012; Pérez-Ahumada et al., 2026).

5 Conclusion

Drawing on the 2024 wave of the OECD Risks That Matter survey across 27 countries, this article has shown that the perceived redistributive justice of the fiscal exchange with the welfare state is anchored in the experience of work, though not in the manner the risk-and-redistribution tradition would predict. Concrete employment insecurity and the perception that others receive undeserved benefits erode perceived justice, and perceiving technology as enabling enhances it; but non-standard employment does not depress justice (platform workers report it more strongly) and perceived technological displacement risk raises rather than lowers it. At the country level, perceived justice is higher in richer countries and, net of that economic composition, lower in conservative regimes than in liberal ones. Above all, universalist welfare institutions appear to decouple perceived redistributive justice from platform work and technological perception, while residual arrangements tie them tightly together; self-employment, by contrast, becomes consequential precisely under universalist arrangements.

Two implications for the comparative study of welfare attitudes follow. First, perceptions and preferences must be kept analytically apart: the same exposure that raises the demand for redistribution may leave the evaluation of one's own exchange unchanged or even more favourable, so findings established for one attitudinal object cannot be transposed to the other ([Roosma et al., 2016](#)). The second is that the comparative purchase of welfare-regime theory on these attitudes lies less in differences of level, largely absorbed by countries' economic composition, than in differences of structuring: regimes shape how strongly individual circumstances translate into judgements of justice. Reframing the regime effect as one of decoupling rather than elevation offers a more accurate account of why, in some institutional settings but not others, the legitimacy of the welfare state fragments along the contours of the labour market.

The findings also bear on the European policy agenda on access to social protection ([Council of the European Union, 2019](#); [European Parliament and Council of the European Union, 2024](#)). That platform workers judge their exchange with the state as comparatively just should not be read as the absence of a problem: on the proportionality reading advanced here, it reflects how little they contribute to and expect from the contributory system. Extending coverage, the aim of the Council Recommendation and, for platform work, of the recent directive, will draw these workers into a fuller exchange in which more is contributed and more is owed, and the legitimacy returns of that extension will depend on whether the ensuing entitlements are proportionate and legible. The self-employment finding adds a caution for the most encompassing systems: formal inclusion that leaves own-account workers at a visible distance from the employee standard may depress, rather than enhance, their sense of a just exchange.

These conclusions are bounded by the design. The data are cross-sectional, so the analysis identifies associations rather than causal effects, and reverse causation is plausible: those who already regard their exchange with the state as just may evaluate technological change more benignly, rather than the reverse. The principal independent variables of theoretical interest rest on single items or short scales, and the platform and self-employment categories are internally heterogeneous, conflating distinct relationships with the protection system under a common label ([Vallas and Schor, 2020](#)). The positive manifold among the technological-perception items signals that the battery does not fully separate the valence of a perception from engagement with technological change as a frame, and the exclusion of the item on replacement by

artificial intelligence, which did not load cleanly on either factor, underscores the point. At the macro level, 27 countries afford limited power for cross-level interactions and place a heavy burden on the regime classification, whose treatment of the post-socialist, southern European and non-European cases, while defensible, remains contestable (Aidukaite, 2011; Ferrera, 1996; Martínez Franzoni, 2008).

These limitations mark out an agenda. Panel data would allow the temporal ordering of technological perception and perceived redistributive justice to be established and reverse causation to be assessed directly. Instrumentation that separates concrete from prospective technological risk, and the valence of a perception from the salience of technological change, would clarify the mechanisms behind the displacement anomaly; qualitative inquiry into how platform and own-account workers construct the reference points against which they judge their exchange would do the same for the platform result. Comparative leverage would be improved by complementing categorical regimes with policy-specific continuous indicators, such as decommodification and active labour-market and social-investment provision, beyond the aggregate social expenditure controlled for here, which would also permit a more direct test of whether policy feedback, rather than reassurance, underlies the institutional patterning observed (Fan et al., 2024; Gallego and Kurer, 2022). Greater attention to within-regime heterogeneity, particularly in the commodified welfare states of Latin America, would test the generalisability of the decoupling thesis beyond the European core (Pérez-Ahumada et al., 2026).

The legitimacy of the welfare state in the digital transition appears to depend less on the objective pace of automation than on whether institutions can prevent perceived redistributive justice from fragmenting along individual labour-market lines. On the evidence presented here, universalist designs are better able to do so. As digitalisation continues to remake the standard employment relationship around which social protection was built (Howcroft et al., 2025), how perceived justice is distributed across the workforce, not only how much redistribution citizens want, is likely to become a central question for the comparative study of welfare legitimacy.

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7 Appendix

7.1 Dimensions of Perceived technological change

```
load(file=here::here("input/proc/digitalization.RData"))

digitalization <- digitalization %>%
  dplyr::select(-tecnoestres_2, -tecnoestres_4)

sjPlot::tab_fa(digitalization)
```

Parallel analysis suggests that the number of factors = 3 and the number of components =

Table 4: Exploratory factor analysis of Perceived technological change items

Table 4: Factor Analysis			
	Factor 1	Factor 2	Factor 3
Replaced by robot	0.76	-0.03	0.10
Replaced by AI	0.30	0.02	0.76
Job replaced by person on internet platform	0.77	0.04	0.00

Lose job due to skills	0.78	-0.01	-0.04
Tech will improve work-life balance	-0.08	0.77	0.04
Tech will make work safer	0.18	0.61	-0.04
Tech will make work less boring/stressful/demanding	-0.03	0.78	-0.00
Cronbach's α	0.83	0.78	

7.2 Country classification and N samples

Table 5: Welfare regime classification

Welfare regime	Country	Original N
Liberal	Canada	1,009
	Chile	1,002
	Estonia	1,001
	Ireland	1,004
	Latvia	1,010
	Lithuania	1,006
	Mexico	1,011
	United Kingdom	1,010
	United States	1,005
Social-democratic	Denmark	1,006
	Finland	1,009
	Netherlands	1,001
	Norway	1,013
	Austria	1,006
	Belgium	1,012

Conservative

Welfare regime	Country	Original N
	France	1,004
	Germany	1,047
	Greece	1,003
	Israel	1,005
	Italy	1,011
	Korea	1,010
	Poland	1,005
	Portugal	1,016
	Slovenia	1,001
	Spain	1,004
	Switzerland	1,009
	Türkiye	1,009
Total	27 countries	27,229

Note: Original N refers to all respondents in the 2024 wave of the Risks That Matter survey; Final N to the analytical sample after listwise deletion of miss